

Part 1 — Where things are

Item	Value
Staff app	https://bwprodsystem.co.za/wages (name + PIN)
Office dashboard	https://bwprodsystem.co.za/admin/wages (any logged-in office user)
Payroll Excel (auditor)	Download Payroll Excel (new) button on the dashboard · /wages-admin/payroll.xlsx?from=YYYY-MM-DD
Old export	Export Excel (old rules – reference only) + CSV — kept until told otherwise
Live code version	v2026-09-23-13 — check at /wages-version
Health	/wages-health
Payroll week	Saturday → Friday
Auditor run	Wednesday. Hours for Wed–Sun are entered as a best guess ("a normal day"); staff correct them the following week.
Database	Cloudflare D1 bw-productions-db
Code	git main, tag restore-2026-09-23-payroll-final
Latest auditor file	docs/BW_Payroll_2026-09-19_to_25_AUDITOR.xlsx — 91 rows, R61 933,86 + Sharleen R3 000 = R64 933,86 , 0 mismatches Excel vs database, 0 open reviews

Part 2 — Pay rules (B&W rate rules v10 — general staff HOURLY BY PLACE, owner 22 Sep 2026; Petrus rule 22 Sep 2026)

General staff are paid **by the hour, to the hour, at the rate of the place they worked.** No fixed day; no before/after premium.

Place (chosen by the work type)	Mon–Sat	Sunday (×1.2)
Warehouse (work type "Warehouse Team")	R81,25 / h (R650 ÷ 8)	R97,50 / h
Venue / event (any other work type)	R95 / h	R114 / h

- **Staff meeting 07:00–07:30, Mon–Fri, is not paid.** A warehouse entry covering that time loses the overlap (max 30 min = R40,63 at R81,25; 15 min if he arrives 07:15). Venue entries are not touched (he is not at the office). Saturday has no meeting. Petrus's R640 set day is not touched. **Owner 23 Sep: the meeting is deducted on public holidays too** (Heritage Day 07:00–15:00 warehouse = 7,5 h, not 8).
- **"Warehouse Team" only means the warehouse when the PLACE says warehouse (owner 23 Sep, Takavaudza rule).** Work type Warehouse Team with venue/area/description naming FNB, Riverside, a stadium, a garden, a client's house etc. is priced by the place named: venue → R95/h, House → gardener R62,50/h. The system cross-checks every Warehouse Team entry against venue, area and description and asks the office warehouse / venue / garden (Part 5b item 4). Example: Takavaudza Sun 20 Sep at the House "Installing lights and garden work", 6 h — owner decided VENUE: 6 × R95 × 1.2 = **R684** (24 Sep).
- A man who moves from the warehouse to a venue is paid the warehouse hours at R81,25 and the venue hours at R95, each as its own entry. Gaps (travel) are not paid. Overlapping entries are counted once.
- Midnight: hours before 00:00 at that day's rate, after 00:00 at the next day's (Sat 22:00–Sun 02:00 = 2 × R95 + 2 × R114 = R418).

Examples: Warehouse Mon 06–16 = 10 h – 0,5 h meeting = 9,5 h × R81,25 = **R771,88** · Warehouse Mon 06–12 = 5,5 h × R81,25 = R446,88 · Warehouse Mon 08–16 (no meeting) = R650 · Warehouse Sat 06–16 = R812,50 · Warehouse Sun 06–14 = 8 h × R97,50 = R780 · Venue Mon 06–16 = R950 · Venue Mon 12–20 = R760 · Venue Sun 08–16 = 8 h × R114 = R912 · Warehouse 06–12 then Venue 12–20 = R446,88 + R760 = R1 206,88.

Owner's reasoning (22 Sep): "R650 ÷ 8 = R81,25. Take the time worked at the warehouse at R81,25, and the difference at R95 if it's at the venue. Stay with an hourly rate, but to the hour. Sundays are the 1.2 rule. 6 to 7 would only be 30 minutes — the 30 minutes for the meeting is not paid, R40,62."

- **Music Bus** (work type *Music Bus*): Mon–Sat **R750 fixed** (07–16) + **R120/h** outside · Sunday **R120/h**.
- **Petrus (Tsotlego Petrus Malakoane) — NEW RULE from 22 Sep 2026 (rate rules v6): set day rate R640,00** for 06:00–16:00, Monday–Saturday. Sunday is never automatic.

Day	Inside 06:00–16:00	Before 06:00 / after 16:00
Monday–Friday	R640,00 set day rate for the complete day, whatever time he arrives (he only starts at 06:00)	R55 / h — not paid automatically. Held and flagged red ("Petrus: time outside 06:00–16:00 on a weekday — approve and set the rate"). Office clicks Approve at R55/h — R... , types another rate , or Not payable — R0 . Only then is it added to the shift and the Excel.
Saturday	R640,00 set day rate	R80 / h , paid automatically
Sunday	Nothing automatic — the whole entry is HELD. Red flag "Petrus: Sunday entry — owner to approve or decline" with the recommendation (R640 set day rate + R80/h outside 06–16). Buttons: Approve Sunday — R... · Other amount · Decline — R0 . R0 is paid until Bernie decides.	(included in the recommendation at R80/h)

Examples: Mon 06:30–16:00 = R640 · Mon 06:30–16:30 = R640 paid + 0,5 h held (R27,50 if approved at R55) · Wed 04:50–16:00 = R640 paid + 1,17 h held · Fri 18:00–22:00 (no day time) = R0 paid + 4 h held · Sat 06:45–13:00 = R640 · **Sat 05:00–17:00 = R800 (day + 2 h × R80)** · Sun 07:00–12:00 = R0 until approved (R640 recommended) · Sun 04:00–18:00 = R0 until approved (R960 recommended) · Sat 20:00–01:00 = R320 paid (Saturday 4 h × R80) + Sunday 1 h held (R80 recommended). Rows paid before 22 Sep under the old Petrus rule (R650 for 07–16 + R81,25/h; Sat/Sun R95/h) are **not** recalculated (closed payrolls are closed). * **Petrus — NO DEDUCTIONS from 22 Sep 2026**. Owner: "All deductions are removed. We no longer deduct anything from him going forward ... unless an additional loan is made, but for now it's zero." The R450 "Rent" weekly deduction (#2) is ended (active = 0, effective to 2026-09-18) and he has **no loans**. Nothing is deducted on the dashboard, Loans & Deductions tab or Auditor Trail. Backup: docs/restore-2026-09-22-petrus/rent_deduction_before.json. * **Gardeners (Givemore, Takavaudza):** House / House-Garden work at **R62,50/h Mon–Sat, Sunday ×1.2 = R75/h** (owner 24 Sep: Takavaudza Sun 20 Sep 6 h = **R450**, not R375 — code fixed v2026-09-24-1). Their Warehouse Team / Team Assistance work is priced as general staff. * **Sharleen Ndlovu:** fixed weekly **R3 000**.

🏠 **Lebo — paid on his own schedule (owner 24 Sep 2026, v2026-09-24-4)**. Owner: "Lebo works differently as he's a student ... keep showing me his wages from Thursday the 24th going forward with a Paid button so I can push it when he's been paid and it disappears. Sometimes weekly, other times fortnightly or monthly." Lebo (staff 16) is registered in wage_own_schedule_staff (from 2026-09-24). The dashboard shows a blue "**Paid on his own schedule**" panel listing every Lebo shift from that date not yet marked paid — **across payroll weeks** — date, times, hours, place, rand, a ✓ **Paid** button per line and ✓ **Paid — all N shifts R...** for the lot, and **TOTAL STILL TO PAY**. Pressing Paid stamps the row (own_schedule_paid_at/by, payroll note "PAID to worker (own schedule) by ... on ...") and the line leaves the list. R0 rows with an undecided review show "held — decide the review first" with no button. Amounts are never changed here (still 🗑 Edit shift / reviews). Route /wages-admin/own-schedule-paid (accepts one id or a comma list). Lebo still appears in the normal per-worker sections and the auditor Excel as before. **Lebo's rate (owner 24 Sep, v2026-09-24-7): R50/h at any place, EVERY clocked hour — NO 07:00–07:30 meeting deduction** (owner: "because the wages are so little ... his rate is far less than anyone else's"). Sunday ×1.2 = **R60/h**, public holiday ×2 = **R100/h** (held for approval). ownerPayKind → student / student_warehouse for staff 16; STUDENT_HOURLY = 50. State on 24 Sep: Thu 24 Sep Heritage 07:00–13:25 (owner: 13:25 is right, the 11:00 confirmation withdrawn and correction line #9864 removed) 6,42 h × R100 = **R641,67** (no meeting deduction) · Fri 25 Sep venue 9 h × R50 = **R450** · Sat 26 Sep venue 06:00–13:00 (owner: 13:00 per the Saturday note, was 12:40) 7 h × R50 = **R350 = R1 441,67 unpaid** (rows were R0 / R855 / R633,33 under the general-staff rule before the correction). Backups docs/restore-2026-09-24-lebo/. **Pay date (owner 24 Sep, v2026-09-24-6): ONE payment on Tue 6 Oct 2026** for everything from 24 Sep (wage_own_schedule_staff.next_pay_date); the panel shows a green "ONE PAYMENT DUE TUE 6 OCT 2026" banner; every clocked hour is paid. To add another own-schedule worker: insert into wage_own_schedule_staff.

- **Midnight rule:** one row; hours before 00:00 at the start day's rate, after 00:00 at the next day's.
- **"Warehouse" only in the wording** (work type not Warehouse): red **Rate-choice** review; Bernie clicks Warehouse (R81,25/h, meeting time unpaid, Sun R97,50) or Venue/Event (R95/h, Sun R114). Also catches the "Wearhouse" misspelling.
- **Never:** R375 half day · R750/R650 fixed days for general staff · Sunday ×1.5 · Music Bus R130 / R190.

Music Bus example: Sat 08–22 = R750 + 6 h × R120 = R1 470 · Sun 07–12 = R600.

Staff: Bhekizitha, Brian, Daniel, Erence, Erick, Isaac, John, Joshua, Patrick, Solomon, Thandanani, Thina — general R90 base. Givemore — House R62,50 / Team Assistance R90. Takavaudza — House-Garden R62,50 / Warehouse Team R90. Lebo — **student R50/h any place** (24 Sep). Petrus — R640 set day rate (see above). Sharleen — fixed weekly.

Sunday ×1.2 and public holiday ×2 apply to EVERY place — confirmed by the owner 24 Sep 2026:

Place	Mon–Sat	Sunday (×1.2)	Public holiday (×2, held for approval)
Garden / House (gardeners)	R62,50/h	R75/h	R125/h
Warehouse	R81,25/h (Mon–Fri meeting 07:00–07:30 unpaid)	R97,50/h	R162,50/h (Mon–Fri meeting deducted first)
Venue / event	R95/h	R114/h	R190/h
Petrus	R640 set day 06–16 (+R55/h held Mon–Fri, +R80/h Sat)	held: R640 + R80/h outside	R160/h every hour, meeting deducted
Lebo (student) — any place	R50/h every clocked hour — no meeting deduction	R60/h	R100/h
Music Bus	R750 fixed 07–16 + R120/h outside	R120/h	R750 fixed 07–16 + R180/h outside
A public holiday that falls on a Sunday is priced as a public holiday (×2), not ×2 × 1.2.			

Public holidays (owner 22 Sep 2026, rate rules v10) — LIVE

Official gov.za list (Public Holidays Act 36 of 1994; a holiday on a Sunday makes the following Monday a holiday) **plus the proclaimed Election Day, Wednesday 4 November 2026**. 2026 remaining: Thu 24 Sep Heritage Day · Wed 4 Nov Election Day · Wed 16 Dec Reconciliation · Fri 25 Dec Christmas · Sat 26 Dec Goodwill. 2027 loaded to Day of Goodwill (observed Mon 27 Dec). The list lives in SA_PUBLIC_HOLIDAYS in src/index.tsx and in the engine table wage_public_holidays (Election Day added 22 Sep).

Work on a public holiday is never paid automatically. Every entry is a **red review** with the recommended amount and three buttons: **Approve public holiday — R...** · **Other amount** · **Decline — R0**. R0 until the owner decides. Recommended amounts:

Who	Public-holiday rate
General staff — Warehouse	R162,50 / h (R81,25 × 2) · Mon–Fri the 07:00–07:30 meeting is deducted first (owner 23 Sep)
General staff — Venue	R190 / h (R95 × 2)
Petrus	No fixed day — R160 / h for every hour (R80 × 2, R80 = R640 ÷ 8)
Music Bus	R750 fixed 07–16 + R180 / h outside 07–16
Gardener House work (Givemore, Takavaudza)	R125 / h (R62,50 × 2) — owner 23 Sep
Sharleen	own rule, unchanged

Examples (Heritage Day): Warehouse 07–15 = 8 h – 0,5 h meeting = 7,5 h × R162,50 = **R1 218,75** · Warehouse 06–16 = 9,5 h × R162,50 = R1 543,75 · Venue 08–18 = R1 900 · Petrus 06:30–14:00 with meeting deducted = 7 h × R160 = **R1 120** · Music Bus 12–18 = R750 + 2 h × R180 = R1 110 · Gardener 08–15 = 7 h × R125 = **R875** · Wed 23 Sep 22:00 → Thu 24 Sep 02:00 = 2 h × R95 paid + Heritage 2 h × R190 = R380 held.

Heritage Day, Thursday 24 September 2026 — what was actually approved (owner 23 Sep). Owner: *"Everyone is in the warehouse Thursday 07:00 to 15:00 — no set-ups. Deduct the meeting. Next week the system must ask me what time they really worked until and flag over- or under-payment."* All Heritage Day rows were set by the office to the planned day (hours in red below are the owner's guess, to be confirmed the following Monday):

Worker	Approved hours	Rule	Amount paid this payroll
Bhekizitha, Daniel, Erence, Erick, Isaac, Joshua, Solomon, Takavaudza, Thandanani	07:00–15:00 warehouse	7,5 h × R162,50 (meeting off)	R1 218,75 each
Patrick	06:30–15:00 warehouse	8 h × R162,50	R1 300,00
Petrus	06:30–14:00 (meeting off)	7 h × R160	R1 120,00 — confirm 14:00 next week
Givemore (gardener)	08:00–15:00 House/Garden	7 h × R125	R875,00 — flag next week if he only worked 4 h
John	07:00–12:00 warehouse + 12:00–18:00 Music Bus	4,5 h × R162,50 = R731,25 + R750 + 2 h × R180 = R1 110	R1 841,25

The planned end **15:00** is recorded in the new table wage_planned_hours (2026-09-24 → 15:00). Any Heritage Day entry a worker captures past 15:00 is flagged red  **CLAIMED PAST 15:00** on the dashboard, and from Monday 28 Sep the dashboard opens a **public-holiday follow-up panel** (Part 4) asking, per worker, "what time did he really work until?" with the rand difference worked out.

Owner 23 Sep, verbatim: "Even if they worked less, I don't want anything to be done. I only want something to be done if we underpaid them." So on Monday: Bernie picks the real end time per person → worked MORE than paid = top-up line at the holiday rule in the 26 Sep payroll; worked less or the same = nothing paid, nothing deducted, answer recorded only. Petrus's holiday pricing also deducts the meeting (06:30–14:00 = 7 h × R160 = R1 120 shows as "as paid").

Owner's words: "Petrus — take the hourly rate, R80, or event rate, and multiply by 2 for the hours worked. No longer a fixed rate on a public holiday. Music Bus fixed 750 for 7 to 4; anything after 4 pm R180 an hour."

Part 3 — Rules for staff capturing

- Calendar allows **last payroll's Saturday** → **this payroll's Friday** only. Older: *"OLDER THAN ONE WEEK — cannot be final-submitted."* Later: *"NEXT PAYROLL — capture it from Saturday."* Server refuses both.
- A previous-week date captured now is a **missed shift**: real date kept, paid in the **current** payroll, pink pill **MISSED SHIFT – actual date**
- Final Shift Check shows the real date; one tick box. **Select all** → **Submit selected** final-submits every ticked draft; locked / out-of-window skipped.
- After Final Submission the card is **Submitted — locked**; locked missed shifts move into **Finally submitted shifts** with *"...do not capture them again."*
- A draft is **R0** until final-submitted.
- Staff never see reviews or the office dashboard.** Nothing the office does can block a worker from capturing.
- Double-tap protection (owner 23 Sep, v2026-09-23-11).** Owner: *"Workers are pressing the button two, three, four times and the same shift lands 2–4 times."* Three layers, none of which can stop a genuine capture:
- In the phone:** the first press locks the form, every button greys out and reads **"Saving... please wait"**, an overlay covers the page; a second press is swallowed. Unlocks after 20 s if the network is slow.
- Saving a new draft** with the **same date + start + end** as a draft he already has, or as a paid row for those hours, is refused with the message *"This shift (Thu 24 Sep 07:00–15:00) is ALREADY SAVED — it was not lost. Open it below and press Final Submission once."* (or *"...has ALREADY been submitted and is in the payroll. It was not entered again. If you worked different hours that day, enter only the different hours."*). Editing an existing draft is never blocked.
- Final Submission** of a draft whose hours are already fully covered by a paid row (this payroll or an earlier one) is refused: *"Not final-submitted: you have ALREADY been paid for ... (shift #..., payroll ...)"*. Every refusal is logged in wage_debug_capture. Tested live on the preview (double-save and double-final-submit both stopped) before deployment.

Stale drafts (owner 22 Sep 2026). A draft the worker saved but never final-submitted, dated **before the capture window** (before last Saturday), can never be submitted and must not stay on the system — it confuses everyone and the day was usually paid already. The dashboard now shows every such draft in **red** — **"STALE DRAFT – ... is before the capture window"** — with a  **Delete draft** button. The office can also delete any other unsubmitted draft that should not be there. Deleting: asks for confirmation, keeps a full copy in the log (wage_debug_capture), voids any open review on it, removes it from the worker's app and the dashboard. A draft that was already final-submitted **cannot** be deleted this way (use the manager correction on the paid shift). 22 Sep: Erick's draft #720 (9 Sep, already paid as shift #9643 R1 200 in the 12 Sep payroll) removed; 14 "ghost" reviews on drafts the workers had already deleted were auto-voided (backups in docs/restore-2026-09-22-stale-draft-720/).

Part 4 — Office dashboard: colours, labels, reviews

What you see	Colour	Meaning
59.00 hours · R5 190,00 by the name	yellow	Claimed — what the system paid at final submission
Admin approved: 59.00 hours · R5 190,00	green	Approved after your decisions · amber <i>"N reviews still open"</i> while undecided
 N open reviews 	orange	Click: this worker's open reviews
REVIEW – ACTION NEEDED	red	Your decision needed (already paid / rate choice / double claim)
REVIEW – open	orange	Manual check (engine overlap review)

What you see	Colour	Meaning
REVIEW – RESOLVED	green	Decided — hours or amount, your name, reason
MISSED SHIFT – actual date ... + OVERLAP/CLEAR	pink / red / green	Previous-week date paid this week, checked against last payroll
7.50 h paid (8.00 clocked) in the hours column	grey	Hours actually paid vs hours clocked when the 07:00–07:30 meeting was deducted (v2026-09-23-5)
 CLAIMED PAST 15:00	red	Entry runs past the end time the owner set for that day in wage_planned_hours (v2026-09-23-7)
Public-holiday follow-up panel at the top of the dashboard	red box	Appears the week after a holiday whose hours the owner set: every holiday row (one line per person, John twice: warehouse + Music Bus), what was paid, a drop-down of real end times (11:00–18:00 in half hours) each showing the rand and <i>UNDERPAID, top-up +R... / worked less, nothing recovered / as paid — nothing to do</i> , Confirm button. Owner rule (23 Sep): only an UNDERPAYMENT creates a top-up line in the current payroll; if he worked less or the same, nothing is done — the answer is only written on the row (HOLIDAY HOURS CONFIRMED). Route /wages-admin/confirm-holiday-hours. Tested live: worked-less → no row, no deduction; worked-more → +R125 top-up (test rows removed).
 Edit shift / Open X's worker app ↗	gold	Manager correction form / worker's own page

Retired 23 Sep (owner instruction — "remove the Pay-rule check, only that"): the *RULE +R... / ⚡ all days match pay rule* panel and tile are gone. Every other flag (already paid, rate choice, crew pattern, overlap, holiday, Petrus, stale draft, planned end) stays.

Approved-hours rule (what the auditor sees): per paid shift, the latest RESOLVED review with approved hours (rate-choice / crew-pattern / Petrus / holiday excluded), else claimed. An **"already paid" review approved as a rand amount** pays exactly that amount on the entry (shown as a correction line). **An owner correction on the row itself ( Edit shift with a reason, or a Heritage Day / difference decision) is final:** the Excel and the green figure show the corrected amount as *"OWNER CORRECTION: ..."* and no review can override it (v2026-09-23-11/-12).

Reviews the system raises

Review	When	Colour	You click
ALREADY PAID — ... (new, 21 Sep)	an entry in the current payroll overlaps a day already paid in an earlier payroll	red	Approve as recommended — R... / Nothing due / change hours
Rate choice: Warehouse or Event/Venue	"warehouse" wording, work type not Warehouse	red	Warehouse / Event-Venue
WAREHOUSE CLAIMED BUT THE ENTRY NAMES "..." (23 Sep)	Warehouse Team selected but venue / area / description names a venue, garden or client	red	WAREHOUSE / VENUE (or the owner's one-line APPROVE when part of the day was paid before)
Public holiday — ... (22 Sep)	any entry on a public holiday	red	Approve public holiday — R... / Other amount / Decline
Manual overlap review (engine)	worker has another entry overlapping this one	orange	Approve hours / Pay as claimed / No issue
Double claim — already paid (engine)	catch-up older than one week	red	0 h

Known quirk (unchanged): the engine sometimes compares a draft with its own final-submitted shift and raises a "manual overlap" review — not a double charge; click *No issue – pay as claimed*.

 **Owner's note panel (v2026-09-24-3, owner 24 Sep).** Any planned end time set for a day in the CURRENT payroll week (wage_planned_hours) shows at the top of the dashboard as a yellow note — *"Sat 26 Sep 2026 — work ended 13:00 (everyone at the warehouse ...)"* — with a live count of entries in so far and how many claim past the time. Entries past it carry the red  **CLAIMED PAST** pill. To add a note for a day, the assistant inserts a row into wage_planned_hours (work_date, planned_end, note, set_by). First use: Sat 26 Sep 2026 → 13:00. Sun 27 Sep 2026 → 11:40 (owner: everyone finished 11:40).

 **Owner DAY RULES (owner 27 Sep 2026, v2026-09-27-1).** Table wage_owner_day_rules(work_date PK, force_kind, staff_off_json, note, set_by). Two things the owner can set for a date: 1. **Forced place** — e.g. Sun 27 Sep: *"all the staff that select warehouse today will use the venue rate — they're building and fixing for a venue event."* force_kind = 'event' → every Warehouse Team / warehouse-wording entry that day is priced as **venue** (R95/h; Sunday R114/h) at Final Submission, on the dashboard green figure and in the Excel; the row's note says "OWNER DAY RULE ...". (force_kind = 'warehouse' does the reverse.) 2. **Staff OFF** — e.g. *"Daniel, Thina and Patrick are off — not allowed to claim Sunday."* staff_off_json = [10,3,13]. An entry from them on that date is **held at R0** at Final Submission with a red review " **You marked X OFF on this day ... If he did work: ... = R...**" and buttons **He did work — approve R...** / Other amount / **He was OFF — decline, R0** (route /wages-admin/petrus-extra, key staff_off|shift:ID). 3. **Per-person START and END time** (v2026-09-27-2/-3; staff_start_json + staff_end_json, e.g. Takavaudza Sun 27 Sep 12:30–14:14 from the Life360 "left 52 The Avenue HOUSE at 14:14" alert) — e.g. *"Takavaudza's Sunday garden shift: he only arrived at 12:30 — can only claim from 12:30 onwards."* staff_start_json = {"2": "12:30"}. If he enters an earlier start, the row is **priced from the owner's time** (his clocked times stay on the row; note "OWNER START ... paid from 12:30 only"; manager_update_reason set so the green figure and Excel keep it) and a red review " **You recorded that X only arrived at 12:30 ... paid from 12:30 (R...).** If he really was there from 10:00 the earlier hours would add R..." offers **OVERRIDE — he was there, add R...** / Other amount / **Keep 12:30 start — nothing more** (key owner_start|shift:ID). Live-tested on the preview (Taka 10:00–15:00 garden Sunday → 2,5 h × R75 = R187,50; override → R375), test rows removed. The  **Owner's notes for this week** panel at the top of the dashboard lists each day rule (forced place, names off, "△ N of them entered a shift anyway") together with the planned-end notes. First use: **Sun 27 Sep 2026 — venue rate for all; Daniel, Thina, Patrick OFF.** Live-tested on the preview with temporary drafts (Daniel → held R0 + review; Bhekizitha warehouse → R684 venue Sunday), test rows removed.

Part 5 — THE "ALREADY PAID" RULE (agreed 21 Sep 2026) — LIVE

Why: wages go to the auditor on Wednesday; Wed–Sun hours are a guess. The next week staff enter what really happened — sometimes different hours, sometimes a different job (Garden → Team, Warehouse → Event, Event → Music Bus). The system must find what was paid, compare it with what is now claimed, and work out **in rand** what is still owed.

What the system does — automatically, on the office dashboard only 1. Looks at every entry in the **current unpaid payroll** (draft or paid), inside the staff capture window (last Saturday → this Friday). 2. Finds any **paid** row for the same worker on the same date (proof — no paid row, no flag). 3. Reads times, venue, area, description and **work type** on both. 4. Prices the new claim at the **rate for the work type the worker selected this week** (weekday / Saturday / Sunday · Warehouse Team · Event/Venue · Music Bus · Petrus · gardener). 5. Splits the answer into **rate correction on the hours already paid + extra hours outside the paid window**, and states **Still due = full day at the correct rule – already paid**. 6. Opens **one red review** with the full story and **one button**.

Example (real — Givemore, Thu 17 Sep 2026)

ALREADY PAID — WRONG RATE + EXTRA HOURS Givemore was paid Thu 17 Sep 2026 08:00–16:00 · Garden (Henley on klip) · House/Garden · "Cleaning" · 8.00 h × R62,50 = R500,00 (payroll 2026-09-12). He now claims 06:00–18:00 · Warehouse (Pretoria and fnb) · Warehouse Team · "Strike down and set up" · 12.00 h for the same day. Work type changed: House/Garden → Warehouse Team (rate R62,50 → R81,25). • Rate correction on the paid hours 08:00–16:00: 8.00 h — worth R650,00 at the correct rule, R500,00 was paid → +R150,00 • Extra hours 06:00–08:00 + 16:00–18:00: 4.00 h → R325,00 **Still due: R475,00** (full day 12.00 h at the correct rule = R975,00 – R500,00 already paid). [**Approve as recommended — R475,00**] [Nothing due — R0,00] ► I don't agree — change the hours

The cases it handles (all verified against the rules)

Last week (guess)	This week (actual)	Result	Button
Garden 08–16 R500	Warehouse Team 06–18	+R150 rate correction, 4 h extra R325	R475,00
Warehouse 07–16 R731,25	Event/Venue 07–16	+R18,75 rate correction	R18,75
Event 07–16 R750	Music Bus 07–20	4 h extra at R120	R480,00
Team 08–16 R650	Garden 08–16	already paid more than the correct rule	R0,00
Event 07–16 R750	same shift entered again	duplicate	R0,00
Event Sat 07–13 R570	Event Sat 07–18	5 h extra at R95	R475,00
Event Wed 07–16 R750	Event Wed 05–19	5 h extra at R90	R450,00

Your options on the review * **Approve as recommended — R...** — one click. Recorded with your name and time. * **Nothing due — R0,00** — for a true duplicate. * **I don't agree — change the hours** — type the extra hours you accept; the system re-prices **live at the same rate it chose** (e.g. system said 4 h → R475; you type 2 → R312,50; 6 → R637,50). Tick/untick *include the rate correction*. Give a reason, click **Approve these hours**. The server recalculates independently and records the working. **You never type a rate**.

What approving does: the approved rand amount is paid **in this payroll** on this entry and appears on the Excel as a correction line ("CORRECTION: already paid R500 earlier; rate correction R150 + extra 4.00 h R325 = approved R475"). The green figure follows. **Last week's paid row is never changed**.

Guard rails: current unpaid payroll only (never a closed/paid one — verified: loading the closed 12 Sep week creates nothing) · only against **paid** rows · office dashboard/Excel only · staff app never involved · flags void themselves if the entry is deleted or corrected · every check wrapped so a failure can't break the page.

Part 5b — CREW PATTERN & VENUE-NAME CHECK (owner 22 Sep 2026) — LIVE

Why: under the hourly-by-place rule the *place* decides the pay (Warehouse R81,25/h · Venue R95/h). "If 90 % say warehouse and 10 % say venue, ask: were they at the venue? They must give a specific name. No name — there's a problem."

What the system does (current unpaid payroll only, on every dashboard load / Excel download, never in the staff app): 1. **Venue must be named — RED**. Any entry priced at the venue rate whose venue field is blank or generic ("venue", "event", "site", "n/a", "x" ...) is flagged: "VENUE NOT NAMED: ... claims the venue rate (R95/h) ... but the venue field is (blank). Ask the worker where he was; if it was the warehouse, re-price at R81,25/h." 2. **Crew pattern — ORANGE**. On any day with 4 or more workers, if the minority place is 25 % or less of the crew, each minority entry is flagged with the names: "CREW PATTERN: 7 of 8 workers say Warehouse on Mon 21 Sep (names) — X says venue 'FNB Stadium'. Was he really there? Confirm the venue, or re-price as Warehouse." And the reverse: "5 of 6 were at a venue (FNB Stadium) — X says Warehouse. Check he was not at the venue (R95/h would be due)." 3. **Decision — two unambiguous buttons, on paid rows AND drafts:** "Where was he? Worker selected 'Warehouse Team' at 'Warehouse'. Your click decides the place and the rate:" **WAREHOUSE — R81,25/h · R650,00 (8 h × R81,25) or VENUE — R95/h · R760,00 (8 h × R95)**. Each button shows the rate and the rand it produces. On a paid row the row is re-priced immediately. On a draft the place is recorded with your name; **when the worker final-submits, the rate you chose is used automatically, whatever work type he picked**, and the paid row's note says "Place decided by office (review #..., name): WAREHOUSE R81,25/h". Nothing is paid before Final Submission. 4. **Venue, Area AND Description are all read — RED (owner 22–23 Sep, v2026-09-22-15 → v2026-09-23-4)**. Owner: "even though he says 'warehouse' underneath, it says 'Pretoria FNB'. You should have picked up that the team was at Pretoria and FNB." and "Givemore writes warehouse then puts a venue name — always cross-check venue, area and description and ask me warehouse / venue / garden." Every Warehouse Team entry is checked: if the venue, the Area box or the description names a real place (FNB, Riverside, a stadium, botanical garden, Pretoria, a client's house ... anything that is not Meyerton / Henley / Randvaal / the warehouse address), it is flagged: "WAREHOUSE CLAIMED BUT THE ENTRY NAMES 'Pretoria and fnb': Givemore selected Warehouse Team at 'Warehouse' ... but wrote 'Pretoria and fnb' — that is a venue, not the warehouse." Same WAREHOUSE / VENUE buttons decide it; when the Area names a venue the **Venue** amount is offered as the default. First live case: Givemore Thu 17 Sep (#10073 → Venue R640 approved). 5. **Part of the day already paid in an EARLIER payroll — the buttons pay only the DIFFERENCE (owner 23 Sep, v2026-09-23-1/-2)**. Owner: "That has already been paid for the missed shift from 8 to 4, so we should only suggest paying the difference ... show me your working out." When the entry overlaps a row paid in a previous payroll, a red box lists what was paid (times, place, hours × rate = rand, shift #, payroll) and each button shows the sum **in the owner's order**: 1. paid hours — what was paid last week (e.g. 08:00–16:00 as House/Garden, 8 h × R62,50 = R500,00) 2. the same hours at the chosen place (8 h × R95 = R760,00) 3. difference (2 – 1) = R260,00 4. additional hours not yet paid at the chosen rate (4 h × R95 = R380,00; Warehouse loses the 07:00–07:30 meeting) **AMOUNT TO APPROVE (3 + 4)** — e.g. Givemore Thu 17 Sep: **Warehouse R434,38 · Venue R640,00. Owner's own layout (v2026-09-23-3)**. Bernie asked for the sum in his words and ONE button. The review now reads, in a white box:

Paid last week 08:00–16:00 · House/Garden · 8,00 h × R62,50 **R500,00** (shift #9705, payroll 12–18 Sep) 06:00–18:00 at a venue 12,00 h × R95 **R1 140,00** Less the Garden already paid – **R500,00 WHAT YOU SHOULD PAY THIS WEEK R640,00 [APPROVE — R640,00]** (or: this was the warehouse — R434,38) The green APPROVE pays the difference on the row (last week's row untouched), writes the sum into the row's payroll note and the Excel breakdown ("DIFFERENCE ONLY

...”), and voids the hour-based overlap / already-paid reviews on the same entry as superseded. Uses the same engine as the Already-paid rule (Part 5), forced to each place. Without an earlier payment the buttons show the full-day amounts as before.
Approved on 23 Sep: Givemore Thu 17 Sep → Venue R640,00 (row #9789). 6. **Excluded from the crew majority:** Petrus, gardener House work, Music Bus, Sharleen (own rules) — but see the next paragraph for gardeners.

Garden entry whose wording sounds like EVENT work — the office is asked (owner 24 Sep 2026, v2026-09-24-2). Owner: “Installing lights in a garden — you should have asked me if that’s a warehouse or a venue, because installing the lights could be a venue or garden work.” Every gardener House / House-Garden entry (Givemore, Takavaudza) is now read for event words — lights/lighting, install, set-up, strike/breakdown, rig, stage, sound, décor, draping, marquee/tent, function, wedding, party, event, tables/chairs, dance floor, festoon/fairy — and flagged red **GARDEN CLAIMED BUT THE WORK SOUNDS LIKE A VENUE JOB** with two buttons: **GARDEN — R62,50/h (Sunday R75)** or **VENUE / FUNCTION — R95/h (Sunday R114)**, each showing the rand. Works on paid rows (re-prices now) and drafts (applied at Final Submission). First case: Takavaudza Sun 20 Sep #9821 “Installing lights and garden work” — **owner decided VENUE: R684,00** (6 h × R95 × 1.2). Paid R375 on 23 Sep → shortfall R309 corrected on the same row. A row that already has a Rate-choice flag is not flagged twice. Decided flags never re-open; flags void themselves when the entry is corrected or deleted.

First live run (22 Sep): one flag — #9955 Takavaudza, Mon 21 Sep, “5 of 6 at a venue (FNB Stadium), this one says Warehouse” — a genuine question for the office.

Part 5c — OVERLAP REVIEWS SHOW THE FULL BREAKDOWN (owner 22 Sep 2026) — LIVE

Owner: “I cannot record a decision if I don’t know if it was already billed. Give me a full breakdown so I know exactly what the hour is that I must approve.”

Every overlap / possible-duplicate review now carries a “**What overlaps what**” box:

- **ALREADY BILLED** — day, date, times, hours, work type, venue, description, **rand amount**, shift number and **which payroll it was paid in** (e.g. “PAID in payroll 2026-09-12 → 2026-09-18 (closed)” or “in THIS payroll, not yet paid”).
- **THIS ENTRY** — the same facts for the entry under review (draft or paid row).
- **Overlap by the clock** — the exact overlapping window and hours (e.g. “06:30–12:00 = 5.50 h overlap”) and “**Not covered by the other entry**” (e.g. “12:00–16:00 (4.00 h) after it”).
- **Recommendation in yellow**, one of:
 - **DIFFERENT DAYS:** ... same clock times, but not the same day — *nothing billed twice. Recommended: pay as claimed.*
 - **Identical times and place** — looks like a **DUPLICATE**. Recommended: approve 0 h (the other one already pays X h).
 - Recommended: approve only the hours NOT already covered — 4.00 h (12:00–16:00). The 5.50 h overlap is already billed on shift #....
 - This is the **SAME** entry (the draft became this paid row) — *nothing billed twice.*

The Approve-hours box, reason and Record decision buttons sit directly under it.

🔁 **Reopen — change a decision (v2026-09-22-15).** Owner: “I need to go back and edit the decided amount to 4 hours. It doesn’t allow me to go back.” Every decided review now shows a small 🔁 **Reopen** button next to “Decided: ...”. Click it → the review goes back to OPEN with the same box and buttons; decide again. The previous decision is kept in the review’s log (decisionHistory) with who reopened it. If the previous decision had added rand to a paid row (Petrus extra / public holiday), that add-on is reversed on the row and noted in the row’s payroll note. Route /wages-admin/reopen-review.

Approved hours are priced by place (v2026-09-22-16). When the office approves fewer hours than claimed (e.g. 4 of 12), the payroll pays those hours at the entry’s own hourly rule — Warehouse R81,25 or Venue R95, honouring any WAREHOUSE / VENUE click on that row — as the last hours of the entry (the extra hours are usually the early/late ones). The Excel and the green “Admin approved” figure agree. Example: Givemore Thu 17 Sep, 4.00 h approved → R325 as Warehouse, R380 if VENUE is clicked on #10073.

Self-compare reviews are auto-voided. The engine sometimes opens an overlap review comparing a draft with the very paid row it became on Final Submission (Patrick #9943 and #9951 were this). The dashboard now voids these automatically with the reason recorded — nothing was billed twice, no decision needed.

Part 6 — Payroll Excel for the auditor (6 tabs)

1. **Wage Detail Linked** (master; yellow = editable) — per shift: claimed / approved / override / **effective** hours and amounts, rate, breakdown, review #, system note, decision.
2. **Auditor Trail Linked** — Employee · Sat...Fri (HR editable · Amount) · **6 MISSED SHIFTS** · **Total Hours (Sat–Fri + Missed)** · Wages · deductions · loans · **NET WAGE**. No Bonus / Gross-incl-bonus. Typing HR re-prices proportionally and flows to Summary.
3. **Auditor Summary** — one line per worker, net, grand total.
4. **Flagged – Reviewed** — every review: flag, note, recommended / claimed / approved, decision, by whom, when, reason. **Each worker in its own light shade**; OPEN pink.
5. **Loans & Deductions** — fixed deductions, loans, balances, repayments.
6. **Missed Shifts** — Worker · shift · **Approved hours** (editable) · Amount · Notes; per-worker shading; feeds block 6 on the Trail. Colour key: yellow editable · light-blue sub-total · pink attention · pastel bands = one worker. Fully formula-linked.

Part 6b — 19–25 Sep payroll: every decision taken with the owner on 23 Sep 2026 (alphabetical)

The owner went through **every worker** before the auditor run. All decisions are recorded on the rows (payroll note / manager reason) and in the reviews; before-states are in docs/restore-2026-09-23-*/.

Worker	Decisions (old → new)
Bhekizitha	Thu 17 Sep FNB 16–18 (missed shift) R190 kept. Heritage Day 07–15 warehouse R1 218,75.
Daniel	Week as claimed (warehouse). Heritage Day R1 218,75. Loan #5 (R3 000 CT scan) marked PAID IN FULL outside the payroll — no wage deductions were or will be made.
Erence	Week as claimed. Heritage Day R1 218,75.

Worker	Decisions (old → new)
Erick	Mon–Wed warehouse (his Mon 21 "FNB" row was really Thu 17 Sep 16:00–18:00 FNB — the same shift as Bhekizitha — moved to the right date, R190 missed shift; a Mon 21 warehouse row was added by the office R690,63). Tue 22 / Wed 23 → warehouse R690,63 each. Heritage Day R1 218,75. New loan #7 R500 (22 Sep).
Givemore	Thu 17 Sep: 08–16 Garden paid R500 last week; now 06–18 at Pretoria/FNB → Venue difference R640 approved (#10073). Heritage Day gardener 08–15 → R875 (flag next week if only 4 h). Earlier 4 duplicate rows already removed 15 Sep.
Isaac	Week as claimed. Heritage Day R1 218,75. Loan #6 R600 deducted in full this payroll as scheduled (owner: "leave it as is"). New loan #8 R500 (22 Sep).
John	Wed 16 Sep: rate corrected to R105/h (owner correction). Thu 17 Sep : duplicate draft #848 deleted; only 16:00–20:00 (4 h × R120) = R480 due (earlier hours paid last week). Mon 21 Sep duplicate → R0 . Heritage Day 07–12 warehouse R731,25 + 12–18 Music Bus R1 110 kept. Week approved R6 461,26 .
Joshua	All shifts listed for the owner. Pastdal Sat 13 Sep → R0 (paid in payroll 12–18 Sep). Wed 23 Sep duplicate → R0 . Heritage Day R1 218,75.
Lebo	Stale draft removed (before capture window).
Patrick	Heritage Day 06:30–15:00 warehouse R1 300 . Earlier Rate-choice #9954 ("Wearhouse — Loading") decided Warehouse.
Petrus	Heritage Day 06:30–14:00, meeting deducted, 7 h × R160 = R1 120 — owner to confirm the 14:00 next week. No deductions (Part 2).
Solomon	Tue 22 Sep → warehouse R690,63 (was venue). Heritage Day R1 218,75.
Takavaudza	Rule set (Part 2): Warehouse Team = warehouse only if the place says warehouse; FNB / Riverside = venue R95; House = garden R62,50. Sun 20 Sep House 6 h = R375 . Heritage Day R1 218,75.
Thandanani	Two unsubmitted drafts (#905, #736) removed. Heritage Day R1 218,75.
Thina	One missed shift 5 h R475 (venue).
Sharleen	Fixed R3 000.

Noise reviews (engine self-compare, superseded hour-based reviews once a rand decision was taken) were voided with the reason recorded. Result sent to the auditors: **0 open reviews, 0 open drafts for 12–25 Sep, Excel = database (0 mismatches)**.

Additional loans (state after 23 Sep)

#	Worker	Taken	Amount	Repayment	Status
1	Erick	8 Aug	R2 000 (car)	R500 / week	R500 outstanding
2	Isaac	11 Aug	R900	R250 / week	R350 outstanding
3	Isaac	16 Aug	R600	R250 / week	R150 outstanding
5	Daniel	23 Aug	R3 000 (CT scan)	—	PAID IN FULL outside payroll (owner 23 Sep)
6	Isaac	1 Sep	R600 (kids' trip)	R600 once	Deducted this payroll (19–25 Sep) — left as scheduled
7	Erick	22 Sep	R500 (asked R600 by WhatsApp 07:15 — "please deduct all this week")	R500 in the 26 Sep – 2 Oct payroll	new
8	Isaac	22 Sep	R500 (asked R600 by WhatsApp 08:15 — fuel/oil)	R250 in 26 Sep – 2 Oct + R250 in 3 – 9 Oct	new

Both new loans carry the worker's WhatsApp wording in the reason and start deducting from the **26 Sep** payroll (deduction_start_date), so they do not touch the payroll now with the auditors. Backups: docs/restore-2026-09-23-loans/.

Part 7 — Everything done 12–23 September 2026

12–14 Sep — working platform baseline, missed-shift tint, Final Shift Check real date, 🐞 Edit buttons, calendar window. **15 Sep** — rate rules v4 + midnight split at Final Submission; week corrected to the rules (14 rows; Patrick Sat 12 Sep R772,50 → R475); rate-choice review; John's Sunday duplicate to red review; **Payroll Excel** 6 tabs; missed shifts in the Excel; auditor sees approved hours only; locked missed shifts under *Finally submitted shifts*; duplicates removed with audit copies (Givemore ×4, Erick ×2); triple-check approved-hours-only passed; Givemore Friday R0 fixed (0 h review voided); 0 mismatches Excel vs database; handover PDFs + restore points. **16 Sep** — Excel HR cells and Missed-Shift approved hours **editable** with auto re-price (v2026-09-16-1); **per-worker shading** (v2026-09-16-2). **18 Sep** — review logic (crew check, both-sides overlap wording, proof rule, duplicate catcher) built, briefly live, **rolled back the same day** because it raised 25 reviews on the already-paid 12–18 Sep payroll. Rows deleted (backup kept), code returned to v2026-09-16-2, reviews table verified identical to the pre-change snapshot. Kept on branch review-logic-2026-09-18, not deployed. Lesson written into the rules: **closed payrolls are closed**. **21 Sep** — **Already-paid rule** (Part 5) built, tested on preview with the real Givemore case and 7 rule cases, deployed as v2026-09-21-1; **change-the-hours** fold-out with live re-pricing added as v2026-09-21-2. First flag: Givemore Thu 17 Sep R475 (open, awaiting Bernie).

22 Sep 2026 (v2026-09-22-1, v2026-09-22-2) * Approved "already paid" amounts on drafts now show immediately in the worker's green figure and row ("pays when final-submitted"), and are not re-flagged after Final Submission. * **Petrus rate rules v6** (table in Part 2): R640 set day rate for 06–16 Mon–Sat; Saturday extras R80/h automatic; **Sunday whole entry held for owner approval** (approve as recommended / other amount / decline); **Mon–Fri time outside 06–16 (R55/h) held for office approval and rate decision** (review key petrus_extra|shift:ID, route /wages-admin/petrus-extra). Earlier the same day the rule was briefly R633,33/day (v2026-09-22-2/-3) — superseded before any Petrus row was paid under it. Applies to every Petrus row final-submitted from 22 Sep; the dashboard rule summary and Excel breakdown text read "R640 set day rate (06–16)". * **Petrus: all deductions removed** — R450 Rent ended, no loans (see Part 2). * **General staff → HOURLY BY PLACE (v2026-09-22-7, rate rules v8)**: Warehouse Team R81,25/h, Venue/Event R95/h, every hour, every day. Replaces every fixed-day rule for general staff (v7's R650/R750 fixed days lasted about an hour and priced no rows). Patrick's 4 venue rows already in the 19–25 Sep payroll re-priced to R95/h (R795 → R902,50; R180 → R190; R90 → R95; R360 → R380; backup docs/restore-2026-09-22-rules-v8/patrick_rows_before.json); his 22 Sep "Wearhouse — Loading" row (work type Normal) got a Rate-choice flag #9954 (Warehouse R771,88 or Venue R902,50; stays R795 until chosen). Closed payrolls untouched. * **Sunday ×1.2 and the unpaid 07:00–07:30 staff meeting (v2026-09-22-8, rate rules v9)**: Warehouse Sun R97,50/h, Venue Sun R114/h; Mon–Fri warehouse entries lose the meeting overlap. Rate-choice #9954 for Patrick's 22 Sep row now reads Warehouse R731,25 (9 h after the meeting) or Venue R902,50. * **Crew pattern & venue-name check (v2026-09-22-9)** — see Part 5b. Module src/crew-pattern.ts, review key crew_pattern|shift:ID / crew_pattern|draft:ID. * **Overlap reviews show the full breakdown; self-compare reviews auto-voided (v2026-09-22-10)** — see Part 5c. Patrick #9943 / #9951

voided (draft vs its own paid row). * **Place decision buttons (v2026-09-22-11)**: every Rate-choice / Crew-pattern review — paid row or draft — offers WAREHOUSE R81,25/h and VENUE R95/h with the rand each produces; a draft decision is applied automatically on Final Submission (owner: "pay as claimed doesn't tell me which one is warehouse and which is venue"). * **Public holidays (v2026-09-22-12, rate rules v10)** — see Part 2 → Public holidays. Also fixed: the closed-week dashboard view had gone blank for one deploy ("too many SQL variables" on the overlap-breakdown lookup) — chunked, verified 12–18 Sep renders with 118 reviews. * **Stale drafts & Delete draft button (v2026-09-22-13/-14)** — see Part 3. Erick #720 removed; ghost reviews auto-voided.

23 Sep 2026 (v2026-09-22-15, v2026-09-22-16) * **🔁 Reopen** on every decided review (Part 5c). **Area-aware crew check** (Part 5b). Overlap box now shows a draft that has since been final-submitted as "final-submitted → shift #... in this payroll (R... on the row)" instead of "not yet final-submitted"; compared drafts show their real hours/amount. * **Approved hours priced by place** on the dashboard and in the Excel; WAREHOUSE / VENUE clicks on crew-pattern flags are honoured in the pricing, and place/Petrus/holiday reviews no longer count as an "approved hours" decision. * **WAREHOUSE / VENUE buttons pay only the difference (v2026-09-23-1/-2)** — see Part 5b item 5. Owner rejected the full-day buttons on Givemore #10073 ("R934,38 / R1 140 is wrong — only suggest the difference"). Type-checker caught a second use-before-declaration bug before production. * **Givemore Thu 17 Sep** (owner instruction): 08:00–16:00 already paid R500 in payroll 12–18 Sep (shift #9705); only 06:00–08:00 + 16:00–18:00 = **4.00 h due**. #9996 reopened and re-decided 4.00 h; #9944 (the earlier R475 = 4 h + R150 rate correction) voided as superseded, old decision kept in its log. Backup docs/restore-2026-09-23-givemore/. Fixed a bug in the area check that would have crashed the dashboard on the first area flag (caught by the type-checker before production).

23 Sep 2026 — payroll day (v2026-09-23-3 → v2026-09-23-12), tag restore-2026-09-23-payroll-final * **v-3 Owner's one-line sum + single APPROVE** on already-paid place reviews (Part 5b item 5): "Paid last week ... / 06:00–18:00 at a venue ... / Less the Garden already paid / WHAT YOU SHOULD PAY THIS WEEK"; Venue offered as default when the Area names a venue. Givemore #10073 approved **R640**. * **v-4 Cross-check every Warehouse Team entry** against venue, area and description (crew-pattern.ts wordingNamesAVenue / venueEvidence; flag WAREHOUSE CLAIMED BUT THE ENTRY NAMES "..."). Final Submission now honours an already-approved "already paid" rand amount on the draft. John Thu 17 Sep: duplicate draft #848 deleted, #9779 corrected to 4 h × R120 = **R480**. * **v-5 Joshua**: Heritage Day approved; Pastdal Sat 13 Sep and Wed 23 Sep duplicate → R0. **Hours column shows paid vs clocked** ("7,50 h paid (8,00 clocked)"). **Final Submission blocked when the hours are already paid**. * **v-6 Holiday rule change (owner)**: warehouse meeting 07:00–07:30 deducted first on public holidays, then ×2. Heritage Day approved for Bhekizitha / Daniel / Erence / Erick; Erick Wed 23 → warehouse. * **v-7 Planned end time**: new table wage_planned_hours (2026-09-24 → 15:00, "everyone in the warehouse, no set-ups"); red 🛑 **CLAIMED PAST 15:00** pill. All Heritage Day rows set to 07:00–15:00 warehouse **R1 218,75** (Patrick 06:30 R1 300; Givemore gardener 08–15 R875; Solomon → warehouse; Erick Mon → warehouse). Backups docs/restore-2026-09-23-holiday/ (shifts_batch1_before, thursday_before_1500, erick_9801_before). * **Erick's mis-dated "Monday FNB"** → Thu 17 Sep 16–18 R190 (matches Bhekizitha); Mon 21 warehouse row entered by the office R690,63. * **John** Wed 16 Sep R105 rate correction; Mon 21 duplicate R0. **Solomon** Tue 22 warehouse R690,63. **Thandanani** drafts #905 / #736 deleted. Noise reviews voided. * **v-11 Pay-rule check UI retired** (owner: only that). **Double-tap protection** — phone lock + overlay, server duplicate-save block, final-submit block (Part 3 item 7). **Public-holiday follow-up panel** + /wages-admin/confirm-holiday-hours (Part 4). Excel: owner corrections final (manager_update_reason → "OWNER CORRECTION"), DIFFERENCE ONLY rows, paid-before R0 no longer applied to a row the owner set to a positive amount. Takavaudza venue / warehouse / Sunday House R375. Petrus Heritage Day 06:30–14:00 R1 120. Lebo stale draft removed. * **v-12 Green "Admin approved"** figure shows owner-corrected rows at the corrected amount (paid hours, R0 rows count 0 h); John #9979 noise review voided. **Verified: Excel = database, 0 mismatches, 0 open reviews** → docs/BW_Payroll_2026-09-19_to_25_AUDITOR.xlsx sent to the auditors (R61 933,86 + Sharleen R3 000 = **R64 933,86**). Backups docs/restore-2026-09-23-final/. * **Loans**: Erick #7 R500 and Isaac #8 R500 (both 22 Sep, WhatsApp wording, deductions start 26 Sep); Daniel #5 marked paid in full; Isaac #6 R600 deducted this week as scheduled (owner: leave as is). Backups docs/restore-2026-09-23-loans/. * **v-13 Follow-up panel**: owner rule "only if we underpaid" — top-up only, never a recovery; half-hour end times 11:00–18:00 with UNDERPAID / worked-less labels; Petrus holiday pricing deducts the meeting. Live-tested both paths, test rows removed, DB back to 91 rows R61 933,86. * Throughout: the type-checker (npx tsc --noEmit) is run before every deploy — it caught two use-before-declaration bugs (hrs in crew-pattern, workRates in index) that would have blanked the dashboard.

Part 8 — Payroll figures

Payroll 19–25 Sep 2026 (sent to auditors Wed 23 Sep) — final as at v2026-09-23-12

Worker	Paid rows (missed shifts)	Hours paid	Amount
Bhekizitha Maphosa	7 (2)	47,00	R4 595,01
Daniel Motaung	5 (0)	44,00	R4 474,38
Erence Mngomezulu	8 (3)	51,00	R4 975,01
Erick Mpho Molefe	6 (1)	46,00	R4 335,64
Givemore Chifetete Kuziwa	7 (1)	57,00	R4 015,00
Isaac Mbele	5 (0)	44,00	R3 981,27
John Simbarashe Mhlanga	10 (2)	75,00	R6 461,26
Joshua Motsamai Nteo	9 (3)	63,50	R4 737,51
Patrick Ngozo	8 (3)	53,50	R5 403,75
Solomon Moyo	7 (2)	51,00	R5 139,38
Takavaudza Chokuda	7 (1)	52,00	R5 019,64 (Sun 20 Sep corrected R375 → R684 venue on 24 Sep)
Thandanani Nkala	5 (0)	44,00	R4 310,01
Thina Dyani	1 (1)	5,00	R475,00
Tsotlego Petrus Malakoane	6 (0)	50,75	R4 320,00
Hourly staff total	91 (19 missed)	684,75	R62 242,86 (was R61 933,86 before the Takavaudza Sunday correction of 24 Sep: +R309)
Sharleen Ndlovu (fixed weekly)			R3 000,00
Gross wages			R65 242,86

Hours are the hours actually paid (meeting deducted where it applies; R0 duplicate rows count 0 h). Heritage Day (Thu 24 Sep) is included at the approved holiday amounts in Part 2. Deductions / loans as per the **Loans & Deductions** tab (Isaac #6 R600 + scheduled instalments on #1–#3; Petrus none; Daniel none). Full detail: docs/BW_Payroll_2026-09-19_to_25_AUDITOR.xlsx.

Fixed (recurring) deductions on the books (wage_recurring_deductions, active):

#	Worker	Type	Weekly	From	Note
3	Erick Mpho Molefe	Car Payment	R1 000	1 Aug 2026	ongoing
1	Isaac Mbele	Eggs	R260	1 Aug 2026	ongoing
5	Joshua Motsamai Nteo	Car Payment	R1 000	26 Sep 2026	Added 24 Sep 2026 on owner instruction. Deducted weekly from wages; the accumulated weekly amounts are paid over to Joshua on the 25th of every month. Ongoing until Bernie ends it (set effective_to). Starts with the 26 Sep – 2 Oct payroll — nothing deducted in 19–25 Sep (Joshua's 19–25 Sep net stays R4 737,51). Backup: docs/restore-2026-09-24-joshua-car/.

Retired: #2 Petrus Rent R450 (ended 18 Sep) · #4 Isaac Car R500 (ended 23 Aug).

To confirm on Monday 28 Sep (follow-up panel): Petrus really finished 14:00? · Givemore worked 08–15 (7 h) or only 4 h? · anybody in the warehouse past 15:00? Only an answer that shows an **underpayment** creates a top-up row in the 26 Sep – 2 Oct payroll; worked-less answers are recorded and nothing is recovered (owner rule).

Payroll 12–18 Sep 2026 (paid 17 Sep) — final figures

111 paid rows (24 missed) · claimed 843,50 h · R72 438,15 · **approved 793,75 h · R71 096,90** · missed 98,00 h · R8 861,25 · deductions R3 210,00 · **net R67 886,90**. Full detail: handover-2026-09-16-complete-hardcopy.pdf.

Part 9 — Restore points

Tag / folder	State
restore-2026-09-23-payroll-final	Live now (v2026-09-23-13) — payroll 19–25 Sep as sent to the auditors; owner sentence + single APPROVE, Warehouse-Team cross-check, double-tap protection, planned end / holiday follow-up, hours column, pay-rule check retired, owner corrections final in Excel. Data backups docs/restore-2026-09-23-{givemore,john,john2,joshua,holiday,final,loans}/.
restore-2026-09-23-difference	v2026-09-23-2 — WAREHOUSE / VENUE buttons pay only the DIFFERENCE when part of the day was paid in an earlier payroll (4-line sum).
restore-2026-09-23-reopen	v2026-09-22-16 — Reopen a decided review, area-aware crew check, partial hours priced by place.
restore-2026-09-22-petrus-v5	v2026-09-22-14 — Petrus v5 rule + rent deduction removed. To undo only the deduction removal: UPDATE wage_recurring_deductions SET active = 1, effective_to = NULL WHERE id = 2
restore-2026-09-21-already-paid	v2026-09-21-2 (already-paid rule, old Petrus rule, rent deduction still active)
restore-2026-09-18-rolled-back	v2026-09-16-2 after the 18 Sep rollback
restore-2026-09-15-approved	15 Sep approved state + full data snapshot (docs/restore-2026-09-15-approved/)
docs/restore-2026-09-21-before-paid-before/reviews.json	Reviews table before the already-paid check first ran (158 rows)
docs/restore-2026-09-16-before-crew-check/reviews.json	Reviews table before any 18 Sep logic (157 rows)
docs/rollback-2026-09-18/removed_new_check_reviews.json	The 25 rows removed on 18 Sep
branch review-logic-2026-09-18	Crew check / duplicate catcher / proof rule — not deployed

```
git checkout restore-2026-09-23-payroll-final
npm run build
npx wrangler pages deploy dist --project-name bw-productions --branch main --commit-dirty=true
curl https://bwprodssystem.co.za/wages-version # expect v2026-09-23-13
```

To go back to before the already-paid rule: git checkout restore-2026-09-18-rolled-back and deploy (expect v2026-09-16-2); any paid_before|... reviews can then be deleted (they are the only rows it writes).

Part 10 — Standing rules (owner's instructions — never to be broken)

- Never break the staff flow.** Staff can always log in and capture; every check runs on the office dashboard only.
- The dashboard stays available to every office user** while work is being done.
- Nothing is ever paid, unpaid or changed automatically.** The system flags, works out and recommends; **Bernie decides**, one click.
- A review must carry proof** — the actual paid entry: times, venue, area, description, work type, hours, rand, payroll week. No proof, no review.
- Already paid → only the difference.** Show what was paid, what is now claimed, what changed (hours and/or work type), and recommend the rand difference: rate correction on paid hours + extra hours.
- The rate comes from the work type the worker selected**, priced by the day rule. Bernie changes **hours**, never types a rate; the system re-prices.
- Closed payrolls are closed.** No new reviews on, and no changes to, a payroll that has been paid. Corrections are paid in the current payroll as a visible line.
- Wednesday rule.** Auditor runs on Wednesday; Wed–Sun hours are estimates and may change the following week — the already-paid check exists for exactly this.

9. **Restore point before anything goes live**; report old → new for any corrected figure; test on the preview with real cases first; show Bernie before deploying.
10. Keep the CSV and old-rules Excel buttons until told otherwise.
11. **"Warehouse Team" only means the warehouse when the place says warehouse** (23 Sep). Venue / area / description naming FNB, Riverside, a stadium, a garden or a house decides the rate — the system asks, Bernie clicks warehouse / venue / garden. **Sunday ×1.2 applies to every hourly rate, gardener House work included (R75/h) — owner 24 Sep.**
12. **The 07:00–07:30 meeting is deducted on public holidays too** (23 Sep) — deduct first, then ×2.
13. **When the owner sets a day's hours for everybody** (e.g. "Thursday everyone in the warehouse 07:00–15:00"), the planned end goes into `wage_planned_hours`; the next week's dashboard must ask what time each man really worked until. **Only an underpayment is corrected (top-up line); if he worked less, nothing is done** — owner 23 Sep. Nothing is adjusted without Bernie's Confirm.
14. **One shift, one row.** Double presses are swallowed on the phone and refused on the server; a shift whose hours are already paid cannot be final-submitted again. A worker who worked different hours enters only the different hours.
15. **An owner correction on a row is final** for the Excel and the green figure — no automatic check may re-price it.

Open items (on Bernie's say-so)

1. **Monday 28 Sep**: answer the public-holiday follow-up panel (Petrus 14:00? Givemore 7 h or 4 h? anyone past 15:00?).
2. Loans #7 (Erick R500) and #8 (Isaac R250 + R250) and Joshua's new fixed Car Payment R1 000/week all start deducting in the 26 Sep – 2 Oct payroll — check the Loans & Deductions tab.
3. Drafts' open reviews not blocking the green figure.
4. Remove CSV / old-rules export after one clean payroll.
5. Push main to GitHub (118 local commits not yet pushed; all tags exist locally).

Done and closed: crew check re-introduced for the current week only (22 Sep) · self-compare reviews auto-voided (22 Sep) · pay-rule panel removed altogether (23 Sep) · Daniel's R3 000 loan confirmed paid in full outside the payroll (23 Sep) · Joshua Car Payment R1 000/week fixed deduction added, first deduction in the 26 Sep – 2 Oct payroll (24 Sep).

Prepared for B&W Productions — Bernie Burness — 23 September 2026 (updated 24 Sep 2026).

Owner notes ON EVERY ROW (v2026-09-27-6, 27 Sep 2026)

Bernie: "I asked you to show notes for me for Sundays... the warehouse doesn't show the message... it doesn't give me the times." Every owner note for a day is printed **on the row itself** (paid rows AND not-final-submitted drafts), not only in the panel at the top: - 📌 forced place (e.g. "everyone at the VENUE rate today — this Warehouse entry is priced as venue, Sunday R114/h"); - 🚫 worker marked OFF that day (held at R0 with red review when final-submitted); - ⌚ per-person window (e.g. Taka 12:30–14:14): green "inside your window ✓", red "starts before / ends after" (trimmed to the window at final submission); - 🚫 entry ENTIRELY outside the per-person window (e.g. Taka draft 931 Warehouse 07:00–12:00 vs 12:30–14:14): **held at R0** at final submission with the staff-off style red review (approve if he did work / decline). Not merely trimmed. - 🕒 general finish time for the day (Sat 26 Sep 13:00, Sun 27 Sep 11:40): green within / red "x h past your time — ask what time he really finished". A per-person window overrides the general finish time for that person. Helper: `ownerNotesForRow()` in `src/index.tsx`; final-submission rule `outsideWindowNote` in `applyOwnerRatesToFinalSubmission`.

Several owner windows per person per day (v2026-09-27-7)

`wage_owner_day_rules.staff_windows_json = [{"<staff_id>":[{"start":"07:00","end":"11:30","place":"Warehouse"}, {"start":"12:30","end":"14:14","place":"House/Garden"}]}`. At final submission an entry is paid for the piece inside a window only (red review to override); entirely outside all windows → held R0. Sun 27 Sep 2026: everyone left the warehouse at 11:30 (planned_end 11:30); Taka warehouse 07:00–11:30 + house 12:30–14:14. Lebo ledger: tick boxes + "Mark the TICKED shifts as PAID"; lines show NOT PAID until ticked. Lebo Heritage Day = 07:00–15:00 R800 (owner 27 Sep).

Office edit form re-prices under the owner rules (v2026-09-28-1)

Bernie 28 Sep: "I'm trying to make changes. It's not allowing me." The engine's Edit Wage Shift form DID save the times, but re-priced the row with its old base formula ($R90/h \times \text{engine Saturday loading} = R840$ for Patrick #9869), so the owner rules were lost and it looked as if the change had not taken. Now after every accepted POST `/admin/wages/shifts/id/edit` the proxy calls `repriceAfterManagerEdit()` (`src/index.tsx`): place from work type / wording / office decision / day rule, Sunday ×1.2, holiday ×2, warehouse meeting, student, gardener; stamps calculation_version 10 and appends "Re-priced after office edit ..." to payroll_note. Rows carrying an owner decision (held at R0, owner-times cap, already-paid difference) are left as saved. The green message after Save now reads "... Priced under your rules: Venue/event: 7 h × R95 = R665,00 (the engine had put R840,00)."

Lebo Sunday ×1.5 (v2026-09-28-2, owner 28 Sep 2026)

Lebo (student, staff 16) Sunday = $R50 \times 1.5 = \mathbf{R75/h}$ — Lebo only; the rest of the team stays on Sunday ×1.2. `STUDENT_SUNDAY_FACTOR = 1.5` in `src/index.tsx`. Sun 27 Sep #9867 re-priced 4,5 h × R75 = R337,50 (was R270). Lebo ledger total due 6 Oct: R1 937,50. The day-rule "everyone at the venue rate" note on Lebo's rows says he stays on his student rate.

Owner amounts 28 Sep 2026 (v2026-09-28-4)

- Givemore #9794 Heritage Day Thu 24 Sep: owner set **R950,00** (formula gave $R875 = 7 \text{ h} \times R62,50 \times 2$). Owner amount — do not re-price.
- Thina (staff 3, fixed weekly) #9876: **R650,00** Heritage Day weekend allowance (owner 28 Sep). Real date 24 Sep, paid in payroll 26 Sep–2 Oct as a manager line with 0 hours, 00:00–00:00. Dashboard shows an "OWNER AMOUNT – for Thu 24 Sep" pill instead of the missed-shift / time checks (`isOwnerAmountLine: hours 0 and start == end` → no planned-end / owner-note checks).

"CATCH-UP" replaces "MISSED SHIFT" everywhere the owner sees it (v2026-09-28-5, owner 28 Sep 2026)

Bernie: "CATCH-UP — use this term, but you always need to explain it, because I can't read your mind." Definition shown with the label every time: a shift the worker did in a PREVIOUS payroll week (already paid out) but only entered afterwards; paid now, in the current payroll, showing the real date worked; checked against what was already paid that day (OVERLAP / CLEAR) so no hour is paid twice. "The day was not missed — the entry was late." Where: admin dashboard legend #bw-catchup-legend, CATCH-UP pill (hover text), per-row "What CATCH-UP means" line on paid rows and drafts, totals ("of which CATCH-UPS"), Excel (column header, sheet 6 title/headers), worker app locked-section heading

and card text. Internal DB marker `missed_previous_week` / description prefix "MISSED SHIFT - actual date ..." is unchanged (the engine relies on it).

Petrus fixed weekly wage — R3 840 guarantee (v2026-09-28-7, owner 28 Sep 2026)

Owner: "It's a fixed day rate, Monday to Saturday. No changes unless I tell you. Public holidays are different, but it doesn't apply to Saturday and Sunday. If there's a public holiday on Saturday/Sunday we take 640×6 and put it in 5 days. You must always clear 3 840. We put it over 6 days for UIF and VAT purposes." - Constants `PETRUS_STAFF_ID = 4`, `PETRUS_DAY = 640`, `PETRUS_WEEK_DAYS = 6`, `PETRUS_WEEK_GUARANTEE = 3840`. - `ownerPayForShift`: for kind 'petrus' a Sat/Sun date is never treated as a public holiday (ordinary R640 Saturday; Sunday held as before). Mon–Fri public holiday keeps the holiday rule (hours – meeting $\times$ R160, held for approval) — that week clears MORE than R3 840. - Dashboard panel `#bw-petrus-week`: the 6 days Mon–Sat of the payroll week, each paid / draft / today / still to come / nothing in; "In this payroll so far" vs the guarantee (R3 840, or $R3\ 840 - R640 +$ holiday amount when a weekday holiday is priced). Button "Top up the missing day(s)" → POST `/wages-admin/petrus-guarantee` adds an owner line (event_name 'Petrus weekly guarantee', 0 h, R640) for each fully-past Mon–Sat day with no paid row and no draft. Only days before today count as missing. - Week 19–25 Sep check: $5 \times R640 +$ Heritage Day R1 120 = R4 320 ✓.

No "Final Shift Check" page — FINAL SUBMISSION submits immediately (v2026-09-28-8, owner 28 Sep 2026)

Owner: "I've told you multiple times we don't want this anymore. When you push final submit, it just finally submits it." The worker's FINAL SUBMISSION button is a GET link to `/wages/drafts/:id/final-check` (engine's confirmation page with tick boxes). The proxy now treats that GET like the confirmation POST: rewrites it to POST `/final-submit` with the engine's three yes-answers (`time_correct` / `end_time_correct` / `information_complete`), runs the normal final-submit handling (Saturday parking, real date, owner-rule re-pricing) and redirects to `/wages/me?submitted=1`. The check page is never shown. Errors go to `/wages/me?error=....` Escape hatch for debugging only: `?bw_show_check=1` still renders the engine page. Tested 29 Sep with a throwaway draft (removed).

Owner corrections 29 Sep 2026 (v2026-09-29-1)

1. Holiday follow-up panel ("REALLY WORKED UNTIL") excludes owner-amount lines (0 h, start = end, e.g. Thina R650) and fixed-weekly staff — those were paid properly and need no re-check.
2. Petrus on a Mon–Fri public holiday: the owner MUST approve (was he actually there?). Review offers two buttons: "Approve public holiday rates — R..." (hours – meeting $\times$ R160) or "Approve fixed rate — R640.00" (choice `fixed_day`), plus other amount / R0.
3. Planned finish time (Sat 26 13:00, Sun 27 11:30 etc.) applies to WAREHOUSE and VENUE staff only — never to Music Bus entries (work type / venue contains "music bus") and never to Petrus. `planEndApplies()` gates both the CLAIMED PAST pill and the on-row note.

Lebo venue rate + Sonop 14:00 (v2026-09-29-3, owner 29 Sep 2026)

Lebo: R50/h is WAREHOUSE ONLY. Venue / collection / setup / event = R75/h (STUDENT_EVENT_HOURLY). Sunday R75/h anywhere. Holiday R100/h. Sat 26 Sep Sonop team (Solomon, Bhekizitha, Daniel, Lebo, Thandanani, Erence) worked to 14:00 — rows re-timed and re-priced; Patrick/Joshua were NOT at Sonop and stay 13:00. A per-person owner window overrides the general planned-end pill. Full detail: `docs/restore-2026-09-29-sonop-lebo/`.

Update 29 Sep (v2026-09-29-4)

- Daniel #9897 Sat 26: NOT on the Sonop team (only 07:00 starter; enters 07:00 on every shift) — reverted to 07:00–13:00 = R570 (owner option B). Sonop 14:00 team is: Solomon, Bhekizitha, Lebo, Thandanani, Erence.
- Lebo rate is read from WHAT HE WRITES (owner: "just read what he writes"): `studentKindFromWording()` — event words (collect / setup / deliver / drop-off / breakdown / strike / standby / activation / event / venue / golf / stadium / hotel / arena / festival / wedding / function / install / rig) → R75, and they WIN over the word "warehouse"; otherwise warehouse words (load / offload / prepar / pack / clean / wash / sort / stock / receiv / warehouse) or Warehouse work type → R50; anything else → R75. No owner question needed.